

BATINGA ANIMAL MEDICAL CENTER

PET APPOINTMENT SYSTEM

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ABSTRACT

The **Batinga Animal Medical Center Pet Appointment System** is a desktop-based system designed to help organize the information needed for pet appointments and clinic records. The system is intended for a veterinary clinic where pet owners provide information about their pets and appointment needs to the admin or staff. Instead of keeping important information only through manual records, the system provides a way for the authorized admin or staff to enter, manage, and store the information. The system mainly focuses on pet appointment management and pet records. The admin or staff is responsible for entering the information provided by pet owners, managing appointment information, and maintaining the official pet and medical records. The user or pet owner does not directly input or edit the important information in the system. The user only provides the necessary information to the admin or staff and may use the account credentials provided to access the available information.

I. PROJECT BACKGROUND

1.1 Client / Organization Background

Batinga Animal Medical Center is a veterinary clinic offering services such as general check-ups, vaccinations, deworming, grooming, and other veterinary care. To manage its operations effectively, the clinic requires a desktop-based application that allows administrative personnel and authorized staff to organize, enter, and maintain essential records, including pet details, appointment schedules, and medical histories. Under this system, pet owners can register their own accounts, provide the necessary information, and view their data, while authorized clinic staff handle all data entry and official record maintenance, ensuring a clear separation of roles that prevents unauthorized access and protects official clinic records.

1.2 Current Process

Under the current manual process, pet owners provide essential details such as the pet's name, breed, age, weight, allergies, medical history, requested services, appointment schedules, and expenses to the clinic staff, who then manually record this information; however, keeping these records separately or on paper leads to inefficient search and update times. The proposed system addresses this inefficiency by allowing users to register their own accounts and enabling authorized admins and staff to digitally enter, store, and quickly retrieve all pet owner provided information within a centralized application, ensuring that staff handle and protect the official clinic records.

II. PROJECT PURPOSE AND OBJECTIVES

2.1 Project Purpose

The purpose of this project is to develop a desktop-based pet appointment system for Batinga Animal Medical Center, providing the admin and staff with an organized platform to manage pet records, appointment schedules, and other critical clinic data. Additionally, the system enables users to register their own accounts and view their available information without granting them authority to enter or alter official records. Ultimately, the project focuses on streamlining information management for the clinic while maintaining strict, controlled access over official clinic data

2.2 Main Project Objective

The main objective of the project is to develop a desktop-based Pet Appointment System that allows users to register their own accounts and authorized admin or staff to manage pet information, appointments, medical records, and other necessary clinic information in an organized manner.

2.3 Specific Objectives

- Allow users to register their own accounts.
- Allow the admin or staff to manage user accounts and system access.
- Allow the admin or staff to manage the information needed for pet appointments.

• Allow the admin or staff to input and update pet information provided by the pet owner.

• Allow the admin or staff to record and manage appointment information.

• Allow the admin or staff to maintain pet medical history and past expenses.

• Allow the admin or staff to manage available veterinary services.

• Allow the admin or staff to organize and store important clinic records.

• Allow pet owners to provide the necessary information about their pets and appointment needs to the admin or staff.

• Allow the system to display relevant appointment and pet information for viewing.

• Provide a simple and organized interface for managing pet appointment records.

• Help reduce confusion and make important pet and appointment information easier for the clinic staff to manage.

3. PROJECT SCOPE AND LIMITATIONS

3.1 Project Scope

ADMIN / AREA	SCOPE
User Account Registration	Users/pet owners can register their own accounts within the system.
Admin Account Management	Admin/staff can manage user accounts and system access.

Pet Information	Admin/staff can enter and manage information provided by the pet owner.
Appointment Management	Admin/staff can enter and manage appointment information based on the information provided by the pet owner.
Pet Record	Admin/staff can maintain medical history and past expense records.
Pet Information Collection	Pet owners provide the necessary information about their pets to the user information so that the admin can view it.
Appointment Information	Pet owners provide their preferred appointment details and service needs to the admin/staff.
Veterinary Services	Admin/staff can manage information about the available veterinary services.
Clinic Records	Admin/staff can organize and maintain important information needed by the clinic.
Information Viewing	Admin/staff can view and review the information stored in the system.

3.2 Project Limitations

- It is designed as a **desktop-based application**.
- The system does not include an online payment gateway.
- The system does not cover complete accounting operations.

- The system does not replace the professional judgment of veterinarians.
- Users cannot directly edit or alter official medical records or other important clinic information.
- Important information must be provided to the admin or staff for proper entry into the system.
- The system depends on the availability of the computer and database used by the clinic.
- Advanced reporting features may not be included if they are outside the project's approved scope.

4. PROJECT FEASIBILITY AND PROJECT CHARTER SUMMARY

4.1 Technical Feasibility

The project is technically feasible because it can be built using simple, commonly available software tools. The application interface will be created using Java GUI to ensure that clinic staff can easily use the system. MySQL will be used as the database to store important records such as pet details, appointment schedules, and medical histories. It will also hold information about user accounts, past expenses, and the various services offered by the clinic. This setup allows authorized staff to organize and view data whenever it is needed. Furthermore, the development team can easily run and test the program on a standard desktop computer. Testing both the Java program and the MySQL database helps the team fix connection and performance issues before the final setup.

4.2 Operational Feasibility

The proposed system is operationally feasible because it is designed around the clinic's core need to organize pet and appointment information through a simple interface for authorized staff, alongside self-service account registration for users. The main responsibility for entering and managing records remains strictly with the admin or staff, while pet owners register their own accounts and provide the necessary details. This setup establishes clear roles and keeps everyone's duties straightforward. By centralizing important information into one application instead of relying completely on separate manual records, the system reduces daily confusion for the clinic.

4.3 Economic Feasibility

The project can be developed with relatively low costs because the development team can use available computers and commonly available software. Java, NetBeans, and database software can be used without purchasing expensive commercial development software. Possible expenses may include transportation, printing, documentation, storage devices, and other project-related needs.

4.4 Project Charter Summary

Charter Item	Sample Entry
Project Title	Batinga Animal Medical Center Pet Appointment System

Project Purpose	To provide an organized, centralized system for managing pet records, appointment schedules, and clinic information while maintaining controlled access over official data.
Primary Users	Admin / Clinic Staff and Pet Owners
Major Deliverable	A desktop-based application developed using Java GUI and MySQL database for managing and viewing pet appointment and record information.
Main Constraint	Strict role separation where pet owners can only view information and provide details, while only authorized admin/staff can enter, edit, and manage official clinic records.
Project Success	Successful deployment of a fully functional, tested desktop application and database that reduces data organization errors and speeds up record management for clinic staff.

5. REQUIREMENTS GATHERING AND ANALYSIS

5.1 Stakeholders

Stakeholder	Role / Interest
Admin/Staff	Responsible for creating accounts and entering, updating, and managing important information.
Pet Owners/Users	Register their own accounts, provide information about their pets and appointment needs, and view information made available to them.
Veterinary Clinic	Benefits from more organized appointment and pet records.
Development Team	Responsible for designing, developing, testing, and documenting the system.

5.2 Requirements Gathering Methods

- **Interview** – ask the clinic staff about the information and processes that need to be included.
- **Observation** – observe how appointment and pet information is currently handled.
- **Discussion** – discuss the proposed system with the client and group members.
- **Document Review** – review available forms or records that show the information normally needed from pet owners.

5.3 Functional Requirements

ID	FUNCTIONAL REQUIREMENTS
FR-1	The system shall allow users (pet owners) to register their own accounts.
FR-2	The system shall allow users to log in using credentials provided by admin/staff.
FR-3	The system shall allow authorized admin/staff to manage user accounts and system access.
FR-4	The system shall allow admin/staff to enter pet information provided by the pet owner.
FR-5	The system shall allow admin/staff to enter and manage appointment information.
FR-6	The system shall allow admin/staff to maintain pet medical history.
FR-7	The system shall allow users to view their available appointment information, past expenses information.
FR-8	The system shall allow users to view their available appointment information.
FR-9	The system shall allow users to view available pet records.
FR-10	The system shall allow users to view their profile information.
FR-11	The system shall allow users to log out.

FR-12	The system shall allow admin/staff to manage important clinic information.
FR-13	The system shall restrict management functions to authorized admin/staff.

5.4 Non-Functional Requirements

CATEGORY	REQUIREMENTS
Usability	The system should have a simple interface that authorized staff can understand and use.
Security	Important management functions should only be accessible to authorized admin/staff.
Reliability	Information entered by the admin/staff should be stored correctly.
Performance	Common functions should respond within a reasonable time on the target computer.
Maintainability	The system should allow authorized staff to maintain records without unnecessary steps.
Accuracy	Information entered into the system should match the information provided by the pet owner or clinic staff.

5.5 Sample User Stories and Acceptance Criteria

USER STORY	ACCEPTANCE CRITERIA
As a pet owner, I want to provide my pet's information so that the clinic can keep a record of my pet.	The user can enter and submit the required pet information, and the information is saved in the system.
As a pet owner, I want to provide my appointment information so that I can schedule my pet's visit.	The user can enter the needed appointment details and submit the request successfully.
As a pet owner, I want to view my pet's information so that I can check if the details are correct.	The user can view the pet information saved under their account.
As an admin, I want to manage pet information so that the clinic can keep organized and updated records.	The admin can view, add, update, and manage pet information in the system.
As an admin, I want to manage appointment information so that the clinic can properly track scheduled visits.	The admin can view, update, and manage appointment information.
As an admin, I want to manage a pet's medical history so that the clinic can keep track of previous records.	The admin can view and update the pet's medical history.

As an admin, I want to manage veterinary services so that users can see the available services.	The admin can add, update, or remove available veterinary services and their information.
As a user, I want to log out of my account so that my information stays protected.	The user can log out successfully and is returned to the login page.

6. DEVELOPMENT METHODOLOGY, SDLC MODEL, AND SOFTWARE TOOLS

6.1 Development Methodology Used

The project will use an **Agile development methodology supported by Scrum**. The project will be divided into smaller development tasks instead of trying to complete the entire system at one time. The team will prioritize important features such as login, account management, pet information, appointment management, and records. The team will develop, test, review, and improve these features during the project.

6.2 SDLC Model Used

The project will use the **Agile SDLC model**. Each part of the system will go through planning, requirements analysis, design, development, testing, and review. Using Agile allows the team to identify problems earlier. If a feature does not work as expected, the team can make changes before continuing to the next part of the system.

6.3 SDLC Model Figure



Figure 1: Agile SDLC Model Used for the Project

6.4 Software and Development Tools Used

SOFTWARE /TECHNOLOGY	USE IN THE PROJECT
Java	Main programming language for the desktop application.
NetBeans	Development environment used to create and edit the Java application.
MySQL / Database	Used to store user accounts, pet information, appointments, and records.

phpMyAdmin	May be used to manage and inspect the database during development.
Github / Backup Storage	Can be used to keep backup copies of project files.

6.5 Reason for Selecting the Methodology and SDLC Model

Agile with Scrum is suitable for the project because the team can work on the system in smaller parts. The team can first develop the important functions and then test them before continuing with additional features.

This approach also allows the team to receive feedback from the client or instructor and make necessary changes. The sample P3 explains that Agile supports short development cycles, testing, feedback, and improvement.

6.6 Expected Development Approach

The development team will:

- Identify and prioritize system requirements.
- Divide requirements into smaller backlog items.
- Design the user interface and database.
- Develop the system by sprint.
- Test completed functions.

- Check the system based on the requirements.
- Present progress to the client/instructor.
- Make approved changes.
- Prepare the final system and documentation.

7. SDLC STAGES AND PROJECT APPLICATION

SDLC Stages	Application to the Batinga Animal Medical Center Appointment Management System
Planning	Identify the clinic's appointment management problems, project objectives, users, scope, and available resources.
Requirements Analysis	Gather the clinic's needs and identify the functional and non-functional requirements of the system.
System Design	Design the system's interface, database, login page, appointment forms, and navigation flow.
Development	Develop the login, appointment recording, appointment management, and pet owner information features using Java and MySQL.
Testing	Test each function, check appointment details, validate inputs, and make sure the system works properly.
Deployment	Install the approved system on the clinic's computer and provide basic instructions for the admin and staff.

Maintenance	Fix errors, update appointment records when needed, back up the database, and improve the system based on the clinic's needs.

Sample note: The SDLC is the step-by-step process we follow to create our system, from planning and designing to developing, testing, and maintaining it. This helps us make sure the system works properly and meets the clinic's needs.

8. PROJECT RISKS MANAGEMENT

8.1 Possible Risks Discovered

The team should identify possible problems that may affect the project's time, budget, features, or quality. These risks should be related to the Batinga Animal Medical Center Appointment Management System.

8.2 Risk Assessment and Response

Risk	Probability	Impact	Response / Mitigation
The clinic changes its requirements during development.	Medium	High	Confirm the clinic's needs first and discuss any changes with the team before making them.

The development takes longer than expected.	Medium	High	Focus on the most important features, check the team's progress every week, and manage the time properly.
Project files or database are lost.	Low	High	Keep backup copies of project files and database records to avoid losing important information.
A team member becomes unavailable.	Medium	Medium	Share tasks, keep records of the work, and assign another member to handle unfinished tasks if needed.
The clinic's computer cannot run the system properly.	Medium	Medium	Test the system on a computer with similar specifications before installing it at the clinic.
Clinic staff find the system confusing to use.	Medium	Medium	Let the staff try the system early and make the screens easier to understand based on their feedback.

9. PROJECT TIME MANAGEMENT AND SCHEDULE

9.1 Major Project Activities

The project schedule shows the tasks that our team needs to complete, how long each task may take, and which tasks should be done first. This will help us finish the Batinga Animal Medical Center Appointment Management System on time.

9.2 Sample Project Schedule

Activity	Duration	Target Period	Main Output
Planning and Client Study	1 week	Week 1	Problems, objectives, and project scope
Requirements Gathering	1 week	Week 2	Confirmed system requirements
System and UI Design	1 week	Weeks 3-4	System layout and database design
Sprint 1 Development	2 weeks	Weeks 5-6	Login, pet owner information, and appointment form
Sprint 2 Development	2 weeks	Weeks 7-8	Saving, editing, and managing appointments

Sprint 3 Development	2 weeks	Weeks 9-10	Appointment status, appointment history, and reports
Testing and Revision	2 weeks	Week 11	Tested system with corrected errors
Deployment and Final Documentation	1 week	Week 12	Installed system and completed documents

9.3 Project Milestone

Milestone	Target
Requirements approved	End of Week 2
Initial design approved	End of Week 4
Core appointment functions working	End of Week 8
All planned features completed	End of Week 10
Testing completed	End of Week 11
Final deployment / presentation	End of Week 12

10. PROJECT BUDGET AND COSTING

10.1 Estimated Project Cost

The budget shows the estimated expenses needed to develop and implement the Batinga Animal Medical Center Appointment Management System. Free tools will be used whenever possible to reduce costs.

Cost Item	Basis	Estimated Cost
Development Software	Free / educational tools	PHP 0
Database Software	Free / open-source option	PHP 0
Transportation	Client meetings and testing visits	PHP 1, 500
Printing / Documentation	Drafts, final copies, forms	PHP 600
Backup Storage / USB	Optional project backup	PHP 500
Hardware Adjustment	Minor peripherals or replacement needs	PHP 1, 500
Contingency	Allowance for unexpected project expenses	PHP 0
TOTAL ESTIMATED COST:		PHP 4, 100

10.2 Cost Justification

The budget focuses on the expenses needed to develop and implement the system, excluding student labor. Transportation is for client meetings and project testing, while printing and documentation are needed for project submission. Backup storage helps keep project files safe,

and the hardware and contingency budgets are intended for unexpected needs during development and deployment.

11. SCRUM / AGILE APPLICATION

11.1 Scrum Roles

Scrum Role	Sample Responsibility
Product Owner	Represents the client's needs, prioritizes features, and checks if the system meets the requirements.
Scrum Master	Organizes Scrum activities, helps solve problems, and keeps the team focused on the sprint goals.
Development Team	Designs, codes, tests, documents, and improves the appointment management system.

11.2 Sample Product Backlog

Priority	Backing Item	Expected Result
High	Admin Login	Authorized staff can access the system
High	Dashboard	Admin can view appointment information
High	Appointment Entry	Staff can record pet and owner details

High	Appointment Scheduling	Staff can set the appointment date and time
High	Save Appointment	Appointment details are saved in the database
Medium	Appointment History	Staff can view previous appointments
Medium	Update Appointment	Staff can edit appointment details
Medium	Appointment Status	Staff can update, confirm, or cancel appointments
Low	Appointment Reports	Admin can view a simple appointment summary.

11.3 Sample Spring Plan

Sprint	Sprint Goal	Main Work
Sprint 1	Create the basic system interface	Admin login, dashboard, and basic navigation.
Sprint 2	Develop the appointment process	Appointment entry, pet and owner details, date and time selection.
Sprint 3	Complete appointment management	Saving, editing, canceling, appointment history, and status updates.

Sprint 4	Prepare the system for final use	Testing, bug fixes, user feedback, and deployment preparation.

11.4 Scrum Events

Scrum Event	How the Team Will Apply It
Sprint Planning	The team will choose the tasks to complete during the sprint and set a clear goal.
Daily Scrum / Team Check-in	Members will share their completed tasks, current work, and problems.
Sprint Review	The team will present the completed system features to the client or instructor and gather feedback.
Sprint Retrospective	The team will discuss what went well, what needs improvement, and how to work better in the next sprint.

11.5 Simple Scrum Progress Tracking

The team will use a task board with three columns: To Do, In Progress, and Done. Each task will be assigned to a team member and updated based on its progress. At the end of every sprint, the team will check the completed tasks and discuss what needs to be improved for the next sprint.

12. PROJECT SUMMARY

The proposed Batinga Animal Medical Center Appointment Management System is designed to make managing pet appointments easier and more organized. The system allows the admin or staff to record appointment details, manage schedules, update appointment statuses, and keep records in one place. Customers can request appointments through phone calls, while the staff handles the recording and management of the information.

The project will use available software and hardware suitable for a student development team. By using Agile and Scrum, the team can develop the system step by step, test its features, and make improvements based on feedback. The project schedule, budget, and risk management plan will also help the team complete the system in an organized way.